241801053- J B Dhejasvi-Transportation Problem

Problem 1-Balanced

```
import pulp

# Problem 1 Data
sources = ['A', 'B', 'C']
destinations = ['K', 'L', 'M']

cost = {
    ('A','K'):2, ('A','L'):3, ('A','M'):1,
    ('B','K'):5, ('B','L'):4, ('B','M'):8,
    ('C','K'):5, ('C','L'):6, ('C','M'):8
}

supply = {'A':30, 'B':40, 'C':20}
demand = {'K':20, 'L':30, 'M':40}

model = pulp.LpProblem("Transportation_1", pulp.LpMinimize)

x = pulp.LpVariable.dicts("ship", cost.keys(), lowBound=0)

model += pulp.lpSum(cost[i,j]*x[i,j] for (i,j) in cost)

for s in sources:
    model += pulp.lpSum(x[s,d] for d in destinations) == supply[s]

for d in destinations:
    model += pulp.lpSum(x[s,d] for s in sources) == demand[d]

model.solve()

print("Minimum Cost =", pulp.value(model.objective))
```

```
Minimum Cost = 330.0
```

Problem 2 (Balanced)

```
import pulp

sources = ['A', 'B', 'C']
destinations = ['K', 'L', 'M']

cost = {
    ('A','K'):4, ('A','L'):6, ('A','M'):8,
    ('B','K'):5, ('B','L'):4, ('B','M'):3,
    ('C','K'):9, ('C','L'):7, ('C','M'):4
}

supply = {'A':20, 'B':30, 'C':25}
demand = {'K':30, 'L':25, 'M':20}

model = pulp.LpProblem("Transportation_2", pulp.LpMinimize)
x = pulp.LpVariable.dicts("ship", cost.keys(), lowBound=0)

model += pulp.lpSum(cost[i,j]*x[i,j] for (i,j) in cost)

for s in sources:
    model += pulp.lpSum(x[s,d] for d in destinations) == supply[s]

for d in destinations:
    model += pulp.lpSum(x[s,d] for s in sources) == demand[d]

model.solve()

print("Minimum Cost =", pulp.value(model.objective))
```

```
Minimum Cost = 325.0
```

Problem 3 (Balanced)

```
import pulp

sources = ['A','B','C']
destinations = ['K','L','M']

cost = {
    ('A','K'):10, ('A','L'):2, ('A','M'):20,
    ('B','K'):5,  ('B','L'):8, ('B','M'):10,
    ('C','K'):7,  ('C','L'):14,('C','M'):6
}

supply = {'A':40,'B':30,'C':20}
demand = {'K':30,'L':30,'M':30}

model = pulp.LpProblem("Transportation_3", pulp.LpMinimize)
x = pulp.LpVariable.dicts("ship", cost.keys(), lowBound=0)

model += pulp.lpSum(cost[i,j]*x[i,j] for (i,j) in cost)

for s in sources:
    model += pulp.lpSum(x[s,d] for d in destinations) == supply[s]

for d in destinations:
    model += pulp.lpSum(x[s,d] for s in sources) == demand[d]

model.solve()

print("Minimum Cost =", pulp.value(model.objective))
```

```
Minimum Cost = 480.0
```

Problem 4 (Balanced)

```
import pulp

sources = ['A','B']
destinations = ['K','L','M','N']

cost = {
    ('A','K'):3, ('A','L'):1, ('A','M'):7, ('A','N'):4,
    ('B','K'):2, ('B','L'):6, ('B','M'):5, ('B','N'):9
}

supply = {'A':50,'B':60}
demand = {'K':30,'L':40,'M':20,'N':20}

model = pulp.LpProblem("Transportation_4", pulp.LpMinimize)
x = pulp.LpVariable.dicts("ship", cost.keys(), lowBound=0)

model += pulp.lpSum(cost[i,j]*x[i,j] for (i,j) in cost)

for s in sources:
    model += pulp.lpSum(x[s,d] for d in destinations) == supply[s]

for d in destinations:
    model += pulp.lpSum(x[s,d] for s in sources) == demand[d]

model.solve()

print("Minimum Cost =", pulp.value(model.objective))
```

```
Minimum Cost = 330.0
```

Problem 5 (Balanced)

```
import pulp

sources = ['A','B','C']
destinations = ['K','L','M']
```

```
cost = {
    ('A','K'):6, ('A','L'):8, ('A','M'):10,
    ('B','K'):7, ('B','L'):11,('B','M'):11,
    ('C','K'):4, ('C','L'):5, ('C','M'):12
}

supply = {'A':35,'B':50,'C':40}
demand = {'K':30,'L':45,'M':50}

model = pulp.LpProblem("Transportation_5", pulp.LpMinimize)
x = pulp.LpVariable.dicts("ship", cost.keys(), lowBound=0)

model += pulp.lpSum(cost[i,j]*x[i,j] for (i,j) in cost)

for s in sources:
    model += pulp.lpSum(x[s,d] for d in destinations) == supply[s]

for d in destinations:
    model += pulp.lpSum(x[s,d] for s in sources) == demand[d]

model.solve()

print("Minimum Cost =", pulp.value(model.objective))
```

```
Minimum Cost = 970.0
```

Problem 6 (Balanced)

```
import pulp

sources = ['A','B','C']
destinations = ['K','L','M','N']

cost = {
    ('A','K'):2, ('A','L'):3, ('A','M'):11,('A','N'):7,
    ('B','K'):1, ('B','L'):0, ('B','M'):6, ('B','N'):1,
    ('C','K'):5, ('C','L'):8, ('C','M'):15,('C','N'):9
}

supply = {'A':6,'B':1,'C':10}
demand = {'K':7,'L':5,'M':3,'N':2}

model = pulp.LpProblem("Transportation_6", pulp.LpMinimize)
x = pulp.LpVariable.dicts("ship", cost.keys(), lowBound=0)

model += pulp.lpSum(cost[i,j]*x[i,j] for (i,j) in cost)

for s in sources:
    model += pulp.lpSum(x[s,d] for d in destinations) == supply[s]

for d in destinations:
    model += pulp.lpSum(x[s,d] for s in sources) == demand[d]

model.solve()

print("Minimum Cost =", pulp.value(model.objective))
```

```
Minimum Cost = 100.0
```

problem 7(unbalanced)

```
import pulp

sources = ['A','B','C','DUMMY']
destinations = ['K','L','M']

cost = {
    ('A','K'):4, ('A','L'):6, ('A','M'):8,
    ('B','K'):5, ('B','L'):4, ('B','M'):3,
    ('C','K'):9, ('C','L'):7, ('C','M'):4,
    ('DUMMY','K'):0, ('DUMMY','L'):0, ('DUMMY','M'):0
}
```

```
supply = {'A':20,'B':30,'C':25,'DUMMY':10}
demand = {'K':30,'L':25,'M':30}

model = pulp.LpProblem("Transportation_7", pulp.LpMinimize)
x = pulp.LpVariable.dicts("ship", cost.keys(), lowBound=0)

model += pulp.lpSum(cost[i,j]*x[i,j] for (i,j) in cost)

for s in sources:
    model += pulp.lpSum(x[s,d] for d in destinations) == supply[s]

for d in destinations:
    model += pulp.lpSum(x[s,d] for s in sources) == demand[d]

model.solve()

print("Minimum Cost =", pulp.value(model.objective))
```

```
Minimum Cost = 295.0
```